

TryHackMe

Careers in Cyber

Learning Notes — The 7 Core Cyber Security Roles

"With over 3.5 million unfilled cyber security positions globally, the field needs you."

1. Why a Career in Cyber Security?

Cyber security is one of the fastest-growing and most in-demand career fields in the world. The industry offers exceptional job security, competitive salaries, intellectually stimulating work, and the opportunity to make a real difference — defending individuals, organisations, and critical infrastructure from harm.

Reason	What It Means	Why It Matters
High Demand	3.5 million+ unfilled positions globally	Job security is essentially guaranteed for skilled professionals.
High Pay	Above-average starting salaries across all roles	Cyber pays well at entry level — and grows significantly with experience.
Exciting Work	From legally hacking systems to hunting threats	No two days are the same; problems are novel and intellectually challenging.
Social Impact	Protecting hospitals, banks, governments, individuals	Cyber professionals defend society's most critical systems.
Diverse Paths	7+ distinct roles, offensive and defensive	Whether you prefer building, breaking, or investigating — there's a role for you.

2. The Cyber Security Landscape

Cyber security roles fall into two broad camps — Offensive (Red Team) and Defensive (Blue Team) — plus specialist and hybrid roles that bridge the two. Understanding this landscape helps you navigate towards the career that fits your strengths and interests.

	Offensive (Red Team)	Defensive (Blue Team)
Focus	Break into systems to find vulnerabilities	Monitor, protect, detect, and respond to threats
Roles	Penetration Tester, Red Teamer	Security Analyst, Security Engineer, Incident Responder, Digital Forensics Examiner, Malware Analyst
Mindset	Attacker — how can I break this?	Defender — how can I stop this?

The 7 Roles

This room covers seven core cyber security career paths: Security Analyst, Security Engineer, Incident Responder, Digital Forensics Examiner, Malware Analyst, Penetration Tester, and Red Teamer. Each is a distinct specialisation with its own skills, tools, and career pathway.

3. The 7 Cyber Security Roles — Deep Dive

3.1 Security Analyst

Security Analyst [Blue Team — Defensive]

Security analysts are integral to constructing security measures across organisations to protect the company from attacks. They explore and evaluate company networks to uncover actionable data and recommendations for engineers to develop preventative measures.

Key Responsibilities

- Work with various stakeholders to analyse the cyber security landscape across the organisation.
- Compile ongoing reports about network safety, documenting security issues and response measures taken.
- Develop security plans incorporating research on new attack tools and trends.
- Monitor SIEM dashboards and triage alerts to identify genuine threats.

Core Skills: SIEM tools (Splunk, Elastic), log analysis, threat intelligence, basic scripting (Python/PowerShell), communication skills.

THM Learning Path: SOC Level 1 — TryHackMe

3.2 Security Engineer

Security Engineer [Blue Team — Defensive / Hybrid]

Security engineers develop and implement security solutions using threats and vulnerability data sourced from the security workforce. They work across web application attacks, network threats, and evolving tactics — with the ultimate goal of retaining security measures to mitigate attack risk.

Key Responsibilities

- Test and screen security measures across software and infrastructure.
- Monitor networks and reports to update systems and mitigate vulnerabilities.
- Identify and implement systems needed for optimal security posture.
- Design and maintain firewalls, IDS/IPS, SIEM, and endpoint protection platforms.

Core Skills: Networking (TCP/IP, protocols), system administration, scripting, cloud security (AWS/Azure), firewall and IDS configuration.

THM Learning Path: Cyber Defense Path — TryHackMe

3.3 Incident Responder

Incident Responder [Blue Team — Defensive]

Incident responders respond productively and efficiently to security breaches. They create plans, policies, and protocols for organisations to enact during and following incidents. This is a high-pressure role — assessments and responses are needed in real time as attacks unfold.

Key Responsibilities

- Develop and maintain incident response plans and playbooks.
- Triage and contain active security incidents quickly to minimise damage.
- Coordinate with IT, legal, and executive teams during major incidents.
- Conduct post-incident reviews and produce lessons-learned reports.

Core Skills: Digital forensics, malware analysis, threat intelligence, log analysis, SIEM, communication under pressure.

THM Learning Path: SOC Level 1 + DFIR Introduction — TryHackMe

3.4 Digital Forensics Examiner

Digital Forensics Examiner [Blue Team — Defensive / Legal]

If you like playing detective, this is the role for you. In law enforcement, the focus is collecting and analysing evidence to charge the guilty and exonerate the innocent. In corporate settings, forensic skills are used to investigate incidents, policy violations, and cyber espionage.

Key Responsibilities

- Collect digital evidence from computers, phones, and networks while observing legal procedures.
- Analyze digital evidence to answer questions related to the case.
- Recover deleted files, examine file systems, and analyse memory images.
- Produce legally-sound forensic reports for court or internal proceedings.

Core Skills: File system forensics, memory analysis, network forensics, chain of custody procedures, tools: Autopsy, Volatility, Wireshark.

THM Learning Path: DFIR Introduction — TryHackMe

3.5 Malware Analyst

Malware Analyst [Blue Team — Specialist]

A malware analyst — also called a reverse engineer — analyses suspicious programs, discovers what they do, and writes reports about their findings. Their core task is converting compiled programs from machine language to readable code, usually in a low-level language. A strong programming background is essential.

Key Responsibilities

- Carry out static analysis of malicious programs by reverse-engineering without execution.
- Conduct dynamic analysis by running malware in a controlled sandbox environment and observing its behavior.
- Document malware capabilities, IOCs (indicators of compromise), and write detection signatures.
- Support incident response teams with malware identification and removal guidance.

Core Skills: Assembly language, C/C++, Python, reverse engineering tools (IDA Pro, Ghidra, x64dbg), sandbox environments (Cuckoo, Any.run).

THM Learning Path: Intro to Malware Analysis — TryHackMe

3.6 Penetration Tester

Penetration Tester [Red Team — Offensive]

Also known as ethical hacking or pentesting, a penetration tester's job is to test the security of systems and software through systemised hacking — uncovering flaws and vulnerabilities. The company then uses these findings to rectify issues before real-world attackers find them.

Key Responsibilities

- Conduct tests on computer systems, networks, and web-based applications.
- Perform security assessments, audits, and analyse policies.
- Evaluate and report on findings, recommending actions for attack prevention.
- Identify and exploit vulnerabilities to evaluate real-world risk in each instance.

Core Skills: Kali Linux, Metasploit, Burp Suite, Nmap, Gobuster, scripting (Python/Bash), report writing.

THM Learning Path: Jr Penetration Tester + Offensive Pentesting — TryHackMe

3.7 Red Teamer

Red Teamer [Red Team — Offensive / Adversarial]

Red teamers share similarities with penetration testers but have a more targeted, adversarial role. While pentesters seek many vulnerabilities across systems, red teamers specifically test the company's detection and response capabilities — mimicking real threat actors with stealth and persistence.

Key Responsibilities

- Emulate the role of a threat actor to uncover exploitable vulnerabilities, maintain access, and avoid detection.
- Assess organisations' security controls, threat intelligence, and incident response procedures.
- Evaluate and report on insights with actionable data to help avoid real-world incidents.
- Engage in campaigns lasting up to a month — typically as an external team for organisations with mature security programs.

Core Skills: Advanced exploitation, lateral movement, evasion techniques, social engineering, C2 frameworks (Cobalt Strike, Sliver), OSINT.

THM Learning Path: Red Teaming Path — TryHackMe

4. Roles at a Glance — Comparison Table

Role	Team	Core Focus	Key Tool(s)	Best For
Security Analyst	Blue	Monitor, triage alerts, report	SIEM / Splunk	Communicators, detail-oriented
Security Engineer	Blue	Build and maintain security systems	Firewalls / IDS	Builders, system architects
Incident Responder	Blue	Contain and recover from attacks	DFIR toolset	Calm under pressure, decisive
Digital Forensics	Blue	Collect and analyse digital evidence	Autopsy / Volatility	Detectives, methodical thinkers
Malware Analyst	Blue	Reverse-engineer malicious software	Ghidra / IDA Pro	Programmers, puzzle-solvers

Penetration Tester	Red	Find vulnerabilities via ethical hacking	Burp Suite / Metasploit	Creative hackers, problem-solvers
Red Teamer	Red	Simulate full adversarial campaigns	C2 Frameworks / OSINT	Strategic thinkers, patient actors

5. Pen Tester vs. Red Teamer — Key Differences

These two offensive roles are often confused. Both involve attacking systems, but their objectives, scope, and methods differ significantly:

Dimension	Penetration Tester	Red Teamer
Primary Goal	Find as many vulnerabilities as possible	Test detection and response capabilities
Scope	Broad — many systems and surfaces	Targeted — specific objectives, adversary simulation
Duration	Days to a few weeks	Weeks to a month or more
Awareness	Security team typically informed	Blue team typically unaware (realistic simulation)
Team	Often individual or small in-house team	Usually an external team
Output	Vulnerability report with remediations	Red team report + Blue team gaps identified
Suited For	All organisations	Mature organisations with established security programs

6. Key Takeaways & Lessons Learned

There Is a Role for Every Type of Mind

Cyber security is not a single job — it is an industry with at least seven distinct career paths. Creative, adversarial thinkers thrive in offensive roles. Methodical, detail-oriented thinkers excel on the defensive side. Programmers find a home in malware analysis. Detectives belong in forensics.

Technical Skills Are Learnable — Mindset Is the Foundation

No one is born knowing how to use Metasploit or Volatility. Every skill in this field is learnable through structured practice. What distinguishes great cyber professionals is curiosity, persistence, and the drive to understand how things work at a deep level.

Report Writing Is As Important As Hacking

Every offensive role — pen testing, red teaming — requires the ability to clearly communicate findings. A vulnerability that cannot be explained to a business stakeholder is a vulnerability that will not get fixed. Writing skills are a genuine career differentiator.

The Talent Shortage Creates Opportunity

With 3.5 million unfilled cyber positions globally, the barrier to entry is lower than in many other technical fields. Employers actively seek career changers and self-taught practitioners. Platforms like TryHackMe provide the hands-on skills that close the gap between interest and employment.

Start Broad, Then Specialise

Begin with foundational topics — networking, Linux, web technologies, scripting — then explore the different roles to find where your interest naturally deepens. Most professionals start in a broad role (Security Analyst, Junior Pen Tester) and specialise over time.

The Cyber Quiz Is a Useful Tool, Not a Final Verdict

The TryHackMe careers quiz can help surface which roles resonate with your personality and preferences. Use it as a starting point for exploration, not a definitive assignment. Many professionals move between roles across their careers.

7. TryHackMe Learning Paths by Role

Role	THM Learning Path	What It Covers
Security Analyst	SOC Level 1	SIEM, log analysis, alert triage, threat intelligence, network traffic analysis.
Security Engineer	Cyber Defense	Network security, endpoint hardening, SIEM, vulnerability management.
Incident Responder	SOC Level 1 + DFIR Intro	Incident lifecycle, forensic analysis, threat hunting, malware identification.
Digital Forensics	DFIR: An Introduction	File system forensics, memory analysis, network forensics, evidence handling.
Malware Analyst	Intro to Malware Analysis	Static/dynamic analysis, reverse engineering basics, sandboxing, IOC extraction.

Penetration Tester	Jr Penetration Tester	Recon, scanning, exploitation, web hacking, privilege escalation, reporting.
Red Teamer	Red Teaming Path	Advanced exploitation, evasion, lateral movement, C2, OPSEC, adversary simulation.

General Starting Point

If you are completely new to cyber security, begin with TryHackMe's Pre-Security path and the Introduction to Cyber Security path before diving into role-specific tracks. These build the networking, Linux, and web fundamentals every role requires.

Summary

Role / Concept	One-Line Summary
Security Analyst	Monitor, triage, and report on the security of an organisation's networks.
Security Engineer	Design and maintain the security infrastructure that protects the organisation.
Incident Responder	Lead real-time containment and recovery from active security breaches.
Digital Forensics	Collect and analyse digital evidence — the detective of the cyber world.
Malware Analyst	Reverse-engineer malicious software to understand, detect, and neutralise it.
Penetration Tester	Ethically hack systems to find vulnerabilities before real attackers do.
Red Teamer	Simulate full adversarial campaigns to test an organisation's detection and response.
3.5M Gap	The global cyber talent shortage makes this one of the best fields to enter today.
Career Quiz	TryHackMe's quiz helps identify which role aligns with your personality and interests.

Source: TryHackMe — Careers in Cyber Room | tryhackme.com